



SportVUE MC1
A helmet with a HUD, with RPM, speed and other info, from Motion Research, USA



Spreadmatic
The spreadmatic is a bread buttering machine that works at industrial speed

Ninja ZX-14

Contrary to popular belief, the most powerful hyperbike on the roads is not the Hayabusa, but the Ninja ZX-14

Geek Life



brake-by-wire works flawlessly when at the limit and the C-ABS performs seamlessly too; not a judder or pulse at lever or brake pedal at all.

Spring roll The suspension

As for the advanced suspension, you'd have to rely on München engineering. If you buy a current-day, K-series motorcycle, you will no longer have to sit around with a C-spanner and a measuring tape, adjusting your suspension before that weekend ride. BMW, who has been known for its radical designs and technological advances, has developed the Electronically Adjustable Suspension or ESA II. Perhaps that should have been EAS, but let's not be trivial about these things after the trouble it has taken to develop the world's first on-the-fly electronically adjustable suspension system

for motorcycles. The ESA can adjust preload, compression and rebound damping (check out the box 'Spring roll' on suspension jargon) and does so by adjusting the spring rate. Adjusting the spring rates was a common practice with purpose-built machines, but involved stripping the suspension and actually changing the springs. BMW engineers got smart here and used a polyetherane bushing (called Elastogran) over the main spring – electronically controlling a small sleeve in the Elastogran – enabling it to compress at different rates as the primary spring compressed against it. This makes the suspension a lot more effective in varying conditions, yet adapts to the rider's comfort levels. In addition to this, the suspension's rebound and compression damping are also adjustable on the fly, which allows the rider to push harder,

SPRING ROLL

We did throw in a lot of jargon when it came to the suspension, here's what we mean when we say –

Preload: the weight of the motorcycle acting on the springs of the suspension without a rider or pillion



Courtesy Ohlins

Compression Damping: the rate at which the suspension compress when under load

Rebound Damping: the rate at which the suspension recoils to come back to its original form

CABLE TIES

Fly-by-wire is now common place on most performance motorcycles. When it comes down to critical systems, motorcycle designers tend to want the computer to do most of the work reducing the human element while riding. The fly-by-wire throttle is one such invention. The twist grip looks just the same and so do the cables that run out of it. But instead of operating the butterflies in the throttle bodies, the cables end up in the ECU. So no matter what positions you've



Credit Yamaha

twisted the throttle to, if the ECU thinks it's inappropriate, it takes over and restricts the throttle action. Sort of like your mum telling you to ride slowly, huh?



and ride faster without having to worry if his suspension is set up to spit him off, or leave his mates in his wake around the next corner.

As we move away from the top class of motorcycles and more towards the commuter end of the spectrum, there are a number of unmentioned black boxes and silicon chips that silently make decisions, save that big spill or even make riding the motorcycles as easy as buttering hot bread. All of which have, at some point of time, been the subject of envy of world champions and local enthusiasts alike. But, unlike ten years ago, when one would have to wait a lifetime to see technology trickle down to everyday motorcycles, it now takes about two years from the first prototype to standard technology in road bikes. It's no secret that smart motorcycles have slowly begun to take away the human element from the top flight of

motorcycle racing and races are won in R&D centres rather than on the race track. And so, while professional riders ask 'How much is too much?', they must realise that unlike the last century, motorcycle racing has come of age, and they're doing it for more than just fun. It's a job, a social responsibility to help develop technology that will someday make motorcycling mundane, yet help save lives.

So Mr. Valentino Rossi, you eight time World Champion you, the next time you decide to tell the world that use of electronic rider aids has killed your enthusiasm to race, make a note that while it may have also killed our enthusiasm to watch you, the fact is you're making the motorcycle world a safer place to be in. More fun? Let me switch the power curve on my motorcycle before I tell you.